Email Campaign Hypothesis Test: Findings Report

Nov 7, 2026 ·

Objective and hypotheses

We tested whether customers who received the Email campaign spend more per order than the Control group, using the 1,200-customer campaign dataset.

- **Null hypothesis (H0):** the mean order value of Email buyers equals the mean order value of Control buyers.
- **Alternative hypothesis (H1):** the mean order value of Email buyers differs from that of Control buyers.
- **Test:** Welch's two-sample t-test (does not assume equal variances), two-sided, at the conventional 0.05 significance level.
- **Who is compared:** buyers only, because customers who did not purchase have an order value of zero by definition.

Data and cleaning

The dataset has 1,200 customers (one row each) and 10 columns, covering campaign group, age, gender, website activity, purchase, order value and satisfaction before and after.

Campaign group	Customers (raw data)
Control	409
Social Media	404
Email	387

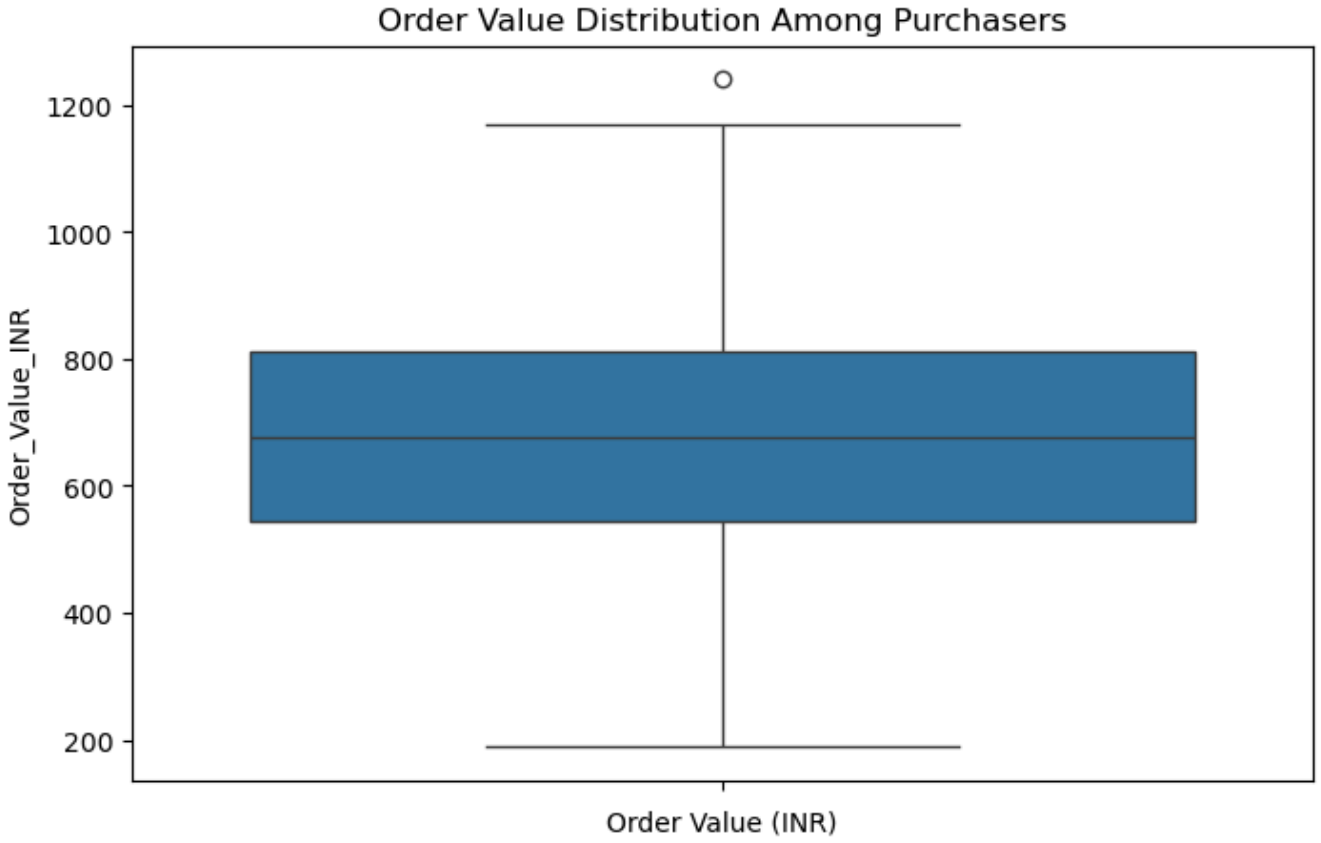
We made three cleaning steps before testing:

1. **Quality checks:** no missing values, no duplicate rows, and every Customer_ID is unique.
2. **Type changes:** order value and session minutes were converted to whole numbers.
3. **Outlier removed:** a boxplot of the 290 buyers (order values from ₹189 to ₹1,241, mean about ₹677) flagged one order of ₹1,241. Orders above ₹1,200 were dropped, leaving 1,199 customers.

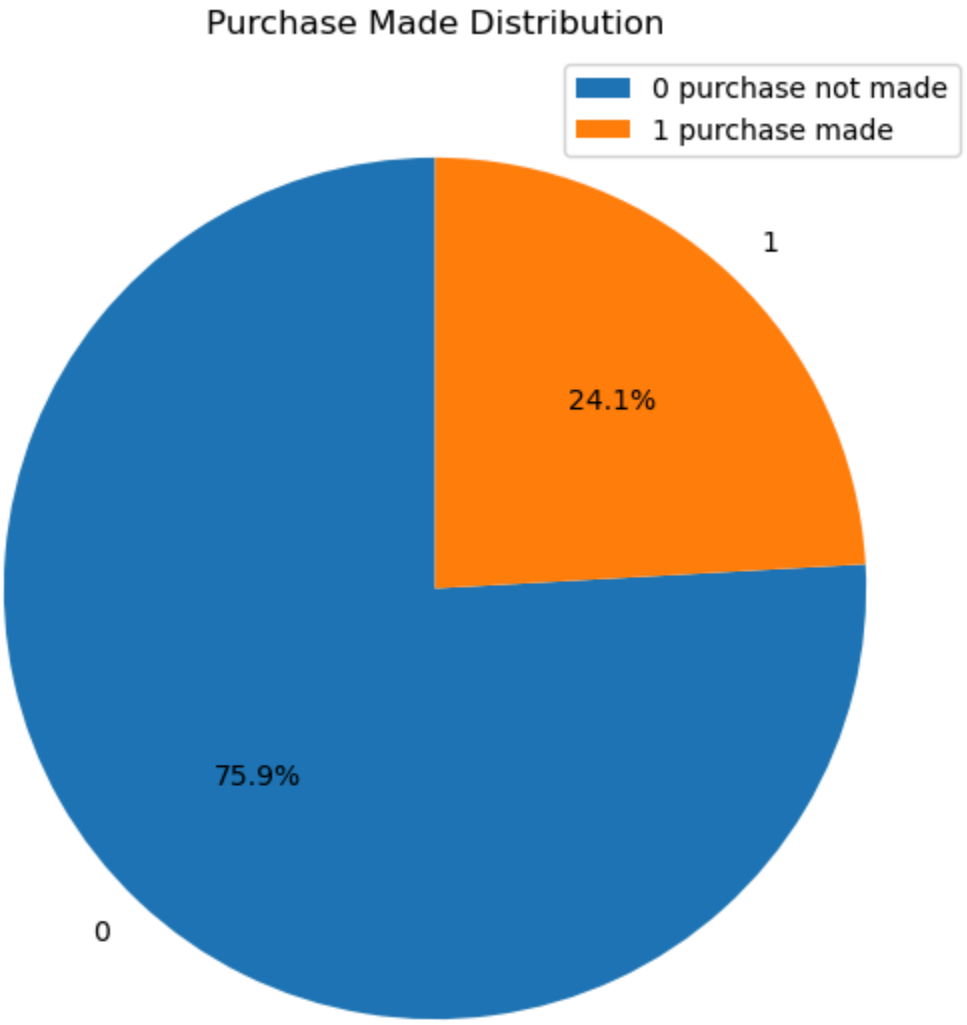
Visualizations

Four charts from the analysis show where the outlier sits, how many customers buy, which group buys most, and how spending compares between Email and Control.

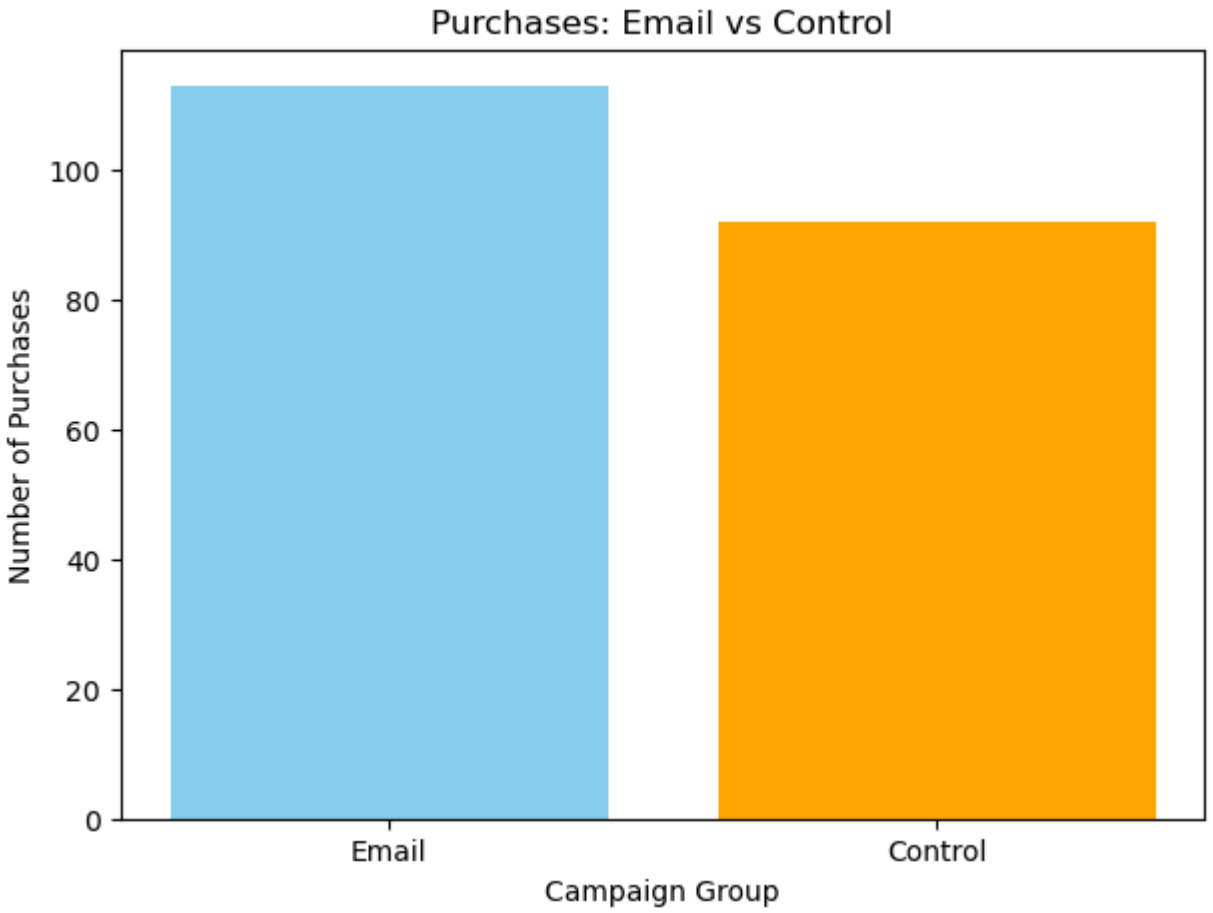
1. Order value among buyers (boxplot). Buyers' orders centre around ₹677, and one order of ₹1,241 sits above the upper whisker, which is why it was removed.



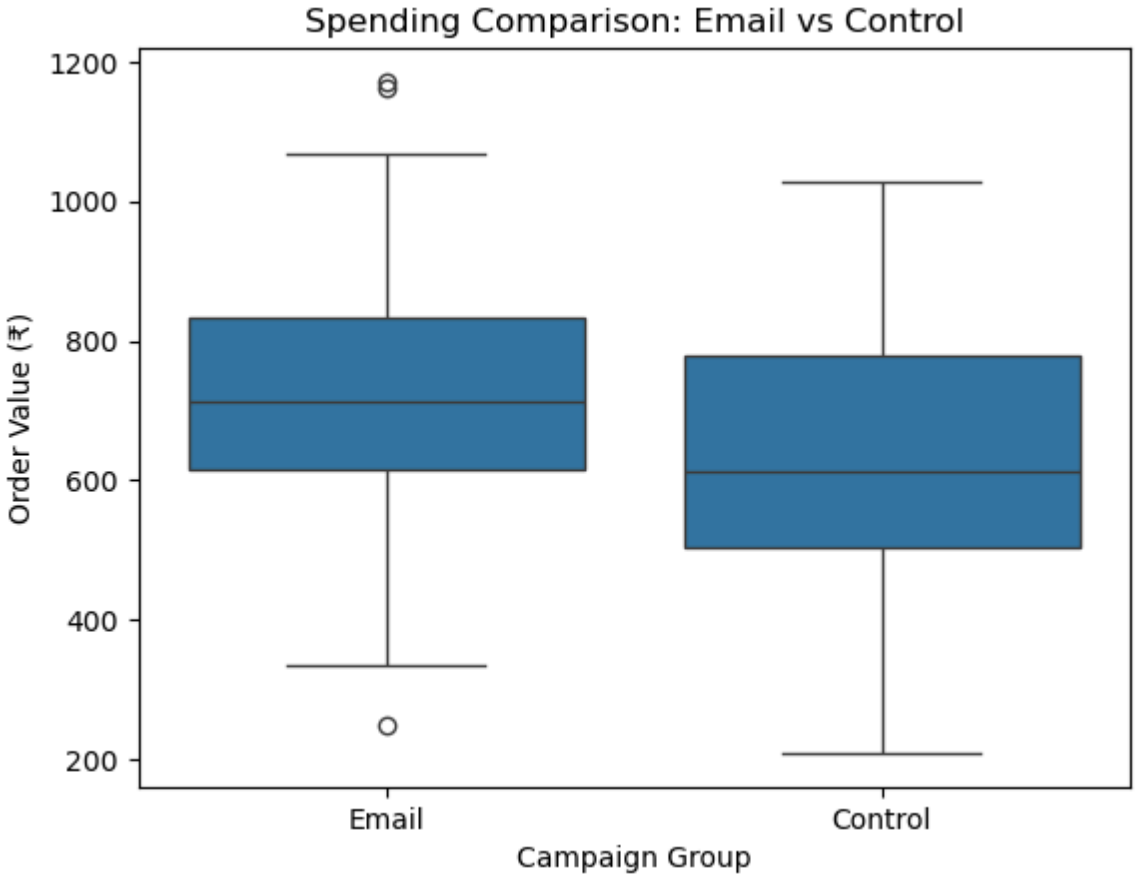
2. Purchase made (pie chart). 24.1% of customers made a purchase and 75.9% did not, so most customers have an order value of zero.



3. Purchases: Email vs Control (bar chart). Email produced 113 buyers against 92 for Control.



4. Spending: Email vs Control (boxplot). The Email box sits higher than the Control box, with a median near ₹710 against about ₹610 for Control.



Hypothesis test

Email buyers spent about ₹83 more per order than Control buyers, and the Welch t-test shows this gap is statistically significant (p = 0.0017, below 0.05), so we reject the null hypothesis.

Group	Buyers	Mean order value (₹)	Std. deviation (₹)
Control	92	633.60	191.43
Email	113	716.23	176.09

- **t-statistic:** 3.19
- **p-value:** 0.0017
- **Difference in means:** ₹82.63, about 13% above the Control mean
- **Decision:** reject H0 at the 0.05 level

What we found

The Email campaign raises how much buyers spend per order: Email buyers average ₹716 against ₹634 for Control, a difference that is very unlikely to be chance.

- **Purchase rate overall:** 24.1% of customers purchased and 75.9% did not (1,199 customers after outlier removal).
- **Buyers by group:** Email had the most buyers (113 from 387 customers), ahead of Control (92 from 409) and Social Media (84 from 404).
- **Order value:** Email buyers spend about 13% more per order than Control buyers, with similar spread in both groups.
- **Significance:** t = 3.19 and p = 0.0017, so we reject the null hypothesis.

Recommendation: keep Email as a campaign channel, and confirm its effect on purchase rate before judging the campaign as a whole. This is based on one test and does not account for campaign cost.